

Day 9 – DevOps Shell Scripting Project: Server Health Check Script

Objective

This day focused on building a real DevOps-style automation script that monitors system health. The script combines concepts learned in previous days such as functions, arrays, loops, conditions, command parsing, logging, and cron automation.

Features Implemented

- Disk usage monitoring with threshold alert
- Service monitoring using pgrep
- Network connectivity checks using ping
- Structured script using functions
- Logging output using tee
- Safe script execution using 'set -euo pipefail'
- Log rotation mechanism for large log files
- Report generation with timestamps

Core Script Structure

```
#!/bin/bash
set -euo pipefail
LOG_FILE="server_health.log"
check_disk()
check_services()
check_connectivity()
log_rotate()
main()
{
    log_rotate
    echo "===== SERVER HEALTH REPORT ====="
    echo "Date: $(date)"
    check_disk
    check_services
    check_connectivity
} | tee -a "$LOG_FILE"
```

Example Output

```
===== SERVER HEALTH REPORT =====
Date: Tue Mar 17 16:05:23 UTC 2026
-----
checking disk usage...
NORMAL: / --> 40% is consumed
checking services status...
nginx is running
docker is running
sshd is running
```

checking network connectivity...
google.com is reachable

Cron Automation

To automate monitoring, the script can be scheduled using cron:
`* /10 * * * * /path/to/health_check.sh`

Key DevOps Concepts Practiced

- System monitoring automation
- Parsing command output with `awk`
- Error-safe scripting with `set -euo pipefail`
- Log management and rotation
- Network connectivity testing
- Service health checks

Conclusion

Day 9 focused on consolidating shell scripting knowledge into a practical DevOps monitoring tool. The resulting script resembles real-world operational automation used in infrastructure monitoring.